

Practice Problem Set 3

ECON 300 – Labour Economics

Winter 2026

1. Competitive Labour Market

Suppose the market labour supply is:

$$N^S = 20W$$

and market labour demand is:

$$N^D = 400 - 10W$$

- (a) Find the competitive equilibrium wage and employment.
- (b) Compute worker surplus and firm surplus.
- (c) Graph the market and clearly label the equilibrium.
- (d) Suppose labour demand increases to:

$$N^D = 500 - 10W$$

Find the new equilibrium and explain the adjustment mechanism.

2. Payroll Tax Incidence

Using the original market in Question 1, suppose a payroll tax of $T = 5$ per worker is imposed on employers.

- (a) Write the new labour demand equation in terms of the wage received by workers.
- (b) Find the new wage paid by firms and the wage received by workers.
- (c) Compute the new level of employment.
- (d) Calculate the tax burden borne by workers and firms.
- (e) Derive the deadweight loss algebraically.

3. Labour Demand: Competitive vs Monopolist Output Market

A competitive firm has:

$$MPP_N = 50 - 2N$$

Output price is $P = 10$.

- (a) Derive the firm's labour demand curve.
- (b) If the wage is 200, find optimal employment.

Now suppose the firm is a monopolist with:

$$P = 20 - Q$$

and production function:

$$Q = 10N$$

- (c) Derive marginal revenue.
- (d) Derive the marginal revenue product of labour.
- (e) Compare employment under monopoly versus competition at wage 200.

4. Basic Monopsony Model

Suppose a monopsonist faces labour supply:

$$W = 10 + 0.5N$$

- (a) Derive the total labour cost function.
- (b) Derive the marginal cost of labour.
- (c) Suppose:

$$VMP_N = 100 - N$$

Find the profit-maximizing employment and wage.

- (d) Compare this outcome to the competitive equilibrium.
- (e) Compute the monopsony wage gap.

5. Minimum Wage under Monopsony

Using Question 4, suppose the government sets a minimum wage of 50.

- (a) Determine whether employment increases or decreases.
- (b) Derive the new marginal cost of labour under the binding minimum wage.
- (c) Determine employment.
- (d) Is this minimum wage efficiency-enhancing? Explain.

6. Perfect Monopsonistic Wage Discrimination

Using the labour supply and VMP_N from Question 4:

- (a) Derive employment under perfect wage discrimination.
- (b) Compare employment under monopsony, perfect discrimination, and competition.
- (c) Compute total worker surplus under discrimination.
- (d) Explain why employment is efficient but distribution differs.

7. Short Run vs Long Run Labour Supply to the Firm

Suppose a firm faces short-run supply:

$$W = 20 + 0.2N$$

and long-run perfectly elastic supply at:

$$W = 40$$

Labour demand is:

$$VMP_N = 100 - 2N$$

- (a) Find short-run equilibrium wage and employment.
- (b) Find long-run equilibrium employment.
- (c) Explain why wage falls in the long run.
- (d) Graph the adjustment path.